Taxonomy

s Worldwide Personal Computing Device Tracker Taxonomy, 2026: Update, 1Q26

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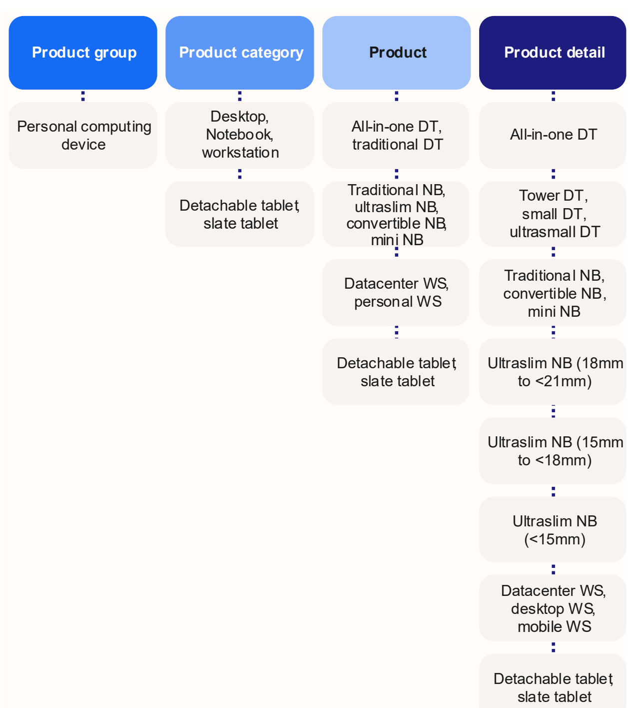
S WORLDWIDE PERSONAL COMPUTING DEVICE TRACKER TAXONOMY

FIGURE 1

Personal computing device roimeasyor egments



Source: IDC, 2026

April 2026, IDC #

PERSONAL COMPUTING DEVICE TRACKER TAXONOMY CHANGES FOR 1Q26

There are no changes.

TAXONOMY OVERVIEW

This taxonomy provides a comprehensive overview of and definitions for personal computing devices (PCDs), which span desktop, notebook, detachable tablet, slate tablet, and workstation product categories. The definitions provided in this document represent ASP scope of IDC s PCD hardware research. In more detail:

This taxonomy outlines market segmentation and measurement methodologies for

the PCD industry, including classification by technology, vendor, geography, and

The taxonomy is used by IDC analysts around the world to generate IDC market

Built on the foundation of IDC s network of regional and country-level research programs, IDC s bottom-up and top-down research delivers a detailed, timely, and accurate view of the market from those closest to it. Quarterly updates and forecasts provide the latest information on rapidly changing markets for time-crucial decision-making.

s PCD research framework and corresponding deliverables are detailed in the Methodology section.

Future outlook

s PCD taxonomy is updated on a regular basis to reflect the changing nature of the marketplace. In addition, related markets and other views of PCD markets may be defined, sized, and forecast as required throughout the year.

Essential uidance

This document, which presents IDC s current taxonomy for the PCD market, should serve as a guide for the existing base of PCD research and for the planned base of IDC hardware research.

IDC expects that the definitions presented in this document will provide explanations for the categories of products that are shown within the data tables of IDC s quantitative deliverables and research documents. Tracker deliverables are usually presented in web query or pivot table format, and the categories are labeled as described in this taxonomy document.

DEFINITIONS

The sections that follow detail IDC s personal computing device (PCD) market definitions.

PCD t

Personal computing devices include desktop, notebook, detachable tablet, slate tablet, and

The following products are specifically excluded from this definition of PCDs:

- **eReaders** are slate form factor devices targeted specifically for reading.
- **Smart displays** are TV or display products that incorporate some embedded functions such as internet access. These products are generally marketed by monitor vendors as displays first, with Android as a complementary feature. But products all - -one (AIO) desktop personal computers (PCs) using an Android interface first, such as HP Inc. s Slate 21, are counted as desktops.
- **PC/TV combinations** that include a full personal computer are counted as desktops.
- **Stick computers** enable computing functionality but require other devices (display, keyboard, etc.) to use them. Although they may functionally support productivity apps desktops and notebooks, tablets, and phones, stick computers are clearly a
- **Smartphones, phablets, or devices** with a display size of less than 7.0in. are not considered PCDs. IDC will reserve the right to classify a device with a screen size of
- **Wearables** generally provide other functions than PCDs. Like phones and thin clients, wearables are covered separately.
- **Application-specific devices** are designed from the start for a dedicated function, such as point-- -sale (POS) terminals, automated teller machines (ATMs), gambling
  - In the case of POS, this would apply to devices specifically designed as POS from
  - solutions that are assembled around a - -
  - are still counted as desktops in IDC
    Device Tracker.
  - Systems designed for digital signage fit under this exclusion.

Other products excluded from this definition of PCDs are (see Table 1):

- Any product, such as a terminal or network computer (NC), that is designed primarily to access information on another computer and that lacks local storage and the ability to operate without being connected to another processor
- Board-level products for building embedded systems or upgrading existing desktop or notebook systems (Raspberry Pi and similar hobby computing platforms fall under this

- Devices for embedded applications
- Upgrades to existing systems (e.g., desktop or notebook computers)
- Single-user RISC-based workstations (e.g., models from Sun Microsystems)

TABLE 1

Examples of xclusions from IDC s orkstation t

		Veriton	Veriton	$EHU Veriton P530
	X	Veriton	Veriton	While Acer +3,QF EliteDesk 800 WSE OK
	EliteDesk 800 WSE	OK	Veriton	The EliteDesk 800 Workstation Edition carries a desktop brand family name of EliteDesk and thus does not qualify as a workstation

Source: IDC, 2025

Technology egmentations

Product category, roedsst, and rocessoal levels US54462026

Table 2 presents PCD product segmentation and taxonomy.

TABLE 2

PCD roUS54462026gory, roodsct, and roUS54462026l levels

Product category	Product	Product dtail
	All- -one DT	All- -one DT
	Traditional DT	Tower DT
		Small DT
		Ultrasmall DT
	Traditional NB	Traditional NB
	Ultraslim NB (18mm to <21mm)	Ultraslim NB (18mm to <21mm)
	Ultraslim NB (15mm to <18mm)	Ultraslim NB (15mm to <18mm)
	Ultraslim NB (<15mm)	Ultraslim NB (<15mm)
	Convertible NB	Convertible NB
	Mini NB	Mini NB
	Personal WS	Desktop WS
		Mobile WS
	Datacenter WS	Datacenter WS
	Detachable tablet	Detachable tablet
	Slate tablet	Slate tablet

Note: The product names in the table reflect actual names in IDC s Worldwide Quarterly Personal Computing Device Tracker database. When used in text (such as analysis reports), these names will often be spelled out (so that DT

Source: IDC s Worldwide Quarterly Personal Computing Device Tracker, 2025

Desktop

Desktops are stationary, AC-powered (no batteries) computing devices. They are not designed to be moved frequently or used on the go. Input is typically via a nonintegrated
- -- y
includes the all- -

Desktop r[illegible]s

All- -one DT. All- -one systems are desktops with a display built into the main computing unit. This category includes tablet-in -
-in - -in
tablet-style display such as the Fujitsu LIFEBOOK GH in which the display is mobile but depends on the base to function.

Immersive PCs desktop systems that integrate a digital entry pad, scanner, and/or projector, along with software integration of these components as a solution are counted under the all- -one DT category. These systems must be sold as a package by the PC vendor, not assembled from individual parts or integrated by a third party.

Traditional DT. Traditional DT includes desktops that are not in the all- -one category.

Desktop product detail levels are as follows:

- **All- -one DT.** These are defined in a previous bullet point.
- **Tower DT.** These designs have a volume of 13.x-L and up from 1Q18 onward.
  - -
- **Small DT.** These designs have a volume of 3 13.x-L from 1Q18 onward. The size
  -
  20.x -20.x-
  8 19.x-L.
- **Ultrasmall DT.** These are also referred to as mini PCs by some in the industry. These designs have a volume smaller than 3.x-L. The size of ultrasmall DTs has diminished over time. Prior to 2013, this category included systems up to 7.x-L. This category includes blade client PCs as well as puck form factors (such as NUC) that are used for personal computing (e.g., excluding those for digital
  - -
- **Blade client.** Blade form factor PCs, sometimes called blade PCs, are rack mountable personal computers that are designed, built, and marketed as single-user systems with a PC operating system (OS). These products are no longer tracked separately but are included under the ultrasmall DT category.

Notebook

A notebook is a portable, battery-powered computing device with a nonremovable keyboard. These devices are designed to be moved frequently and used in a mobile environment. Input is typically keyboard and mouse first; touch-enabled display may be an alternate input method. Designs are generally clamshell, slider, or twister, but other designs are not automatically excluded. This category includes the traditional NB, ultraslim NB delineated along millimeter-based groups, convertible NB, and mini NB.

Notebook products

Traditional NB. This is any notebook not covered in another category.

Ultraslim NB (18mm to <21mm). This notebook PC is at least 18mm thick but also less than 21mm thick at its largest point and is not covered in the convertible NB, mobile WS, or mini NB categories. The ultraslim category includes most products s Ultrabook criteria but also includes non-Intel-based products and Intel products that are thin but may not meet another Ultrabook criteria. For data prior to 1Q17, all ultraslim notebooks less than 21mm at its largest point are placed in this product group (18mm to <21mm), even if it is less than 18mm thick. For data prior to 1Q17, notebooks measuring 21mm or thicker are placed in traditional NB, while <21mm are placed within ultraslim NB (18mm <21mm).

Ultraslim NB (15mm <18mm). This notebook PC is at least 15mm but also less mm thick at its largest point and is not covered in the convertible NB, mobile WS, or mini NB categories. This product value takes effect from 1Q17 onward.

Ultraslim NB (<15mm). This notebook PC is less than 15mm thick at its largest point and is not covered in the convertible NB, mobile WS, or mini NB categories. This product value takes effect from 1Q17 onward.

Convertible This notebook computer is equipped with an integrated keyboard and display that can be used in either a traditional notebook configuration (open clamshell with keyboard accessible) or a slate configuration (flat with no keyboard showing) or includes tabletlike devices that have a foldable configuration (clamshell foldable). The convertible function is often achieved with a rotating or swiveling hinge, - -out or unfold, so that the screen appears exposed on the exterior of both sides or open and close the device similar to a book with the screen within. These devices can use a pen or a similar pointing device to navigate or input data into the -like configuration both less than 2 are s Flex systems that open more than a traditional clamshell design but not far enough to convert to a flat slate

Convertible and ultraslim.

Mini This notebook computer has a screen measuring 7 12in. diagonally, using -performance central processing unit (CPU) (such as an Intel Atom or VIA CPU) that was marketed as a netbook or ultra-low-cost notebook (ULCNB) from 2007 to 2014.

Note: Mini NB devices were introduced in late 2007 with the Eee PC from ASUS and grew very quickly across regions and vendor offerings. However, their fortunes quickly turned with the introduction of tablets as well as other low-cost computers to the extent of damaging the netbook brand. Lacking consumer demand, vendors have updated their product offering and repositioned their low-cost systems away from the netbook/mini notebook category.

Workstation

Workstations are client computers that are specifically designed and configured to meet technical computing requirements. Workstations are counted in our workstation tracker provided they belong to a brand that is dedicated to and known for workstations. S s Precision, HP

s Precision, HP Inc, s Z, and Lenovo s

Premium workstations

Premium workstations provide best- -

These systems are all ISV certified and can typically be configured up to Intel s Xeon

Performance workstations -

graphics but boast less potential horsepower and scalability than their premium

Value workstations are new additions to the workstation tracker. Some of these devices utilize consumer-facing graphics cards and are thus not ISV certified. Models in this class should be thought of as being a workstation predominantly in branding only.

TABLE 3

Workstation tiers

Product dtail	WS branding	ISV certification	Professional raphics	ECC memory	DIMM lots	Native drive bays	WS tier
Desktop WS (including rack and datacenter WS)	Yes	Yes	Yes	Yes		4	Premium
	Yes	Yes	Yes	Yes	2	2	Performance
	Yes	No	No	No	Any	Any	Value
Mobile WS	Yes	Yes	Yes	Yes	4	3	Premium
	Yes	Yes	Yes	No	2	1	Performance
	Yes	No	No	No	Any	Any	Value

Source: IDC, 2025

Slate tablet

Slate tablets are portable, battery-powered computing devices that do not have a permanently attached keyboard and use touch as a primary user interface (though this may . . Both types of tablets may use LCD or OLED displays (epaper-based ereaders are not included here). The device must have a color display that is at least 7.0in. but also less than 16.0in. Slate tablets do not offer an optional first-party keyboard.

Additional notes are as follows:

Applications. Slate and detachable tablets support a broad range of applications, which differentiates them from single purpose focused devices such as ereaders,

Cellular. Tablets may support cellular voice connectivity as a secondary feature.

Exceptions to the rule. IDC will reserve the right to classify a device with a screen size of 0.1in. to smaller than 7in. as a tablet should it meet analyst expectations of a tablet.

Smartphone. Devices with both cellular connectivity and a screen size of less than 7.0in. are not included in this taxonomy but are instead captured in IDC s Worldwide

Quarterly Mobile Phone Tracker. This also includes phablets, which are devices with

All in one. Battery-powered slate devices without permanently attached keyboards and a screen size of 16in. or greater are defined as all- -one desktops.

Convertibles. Devices with a permanently attached keyboard that can flip or spin around to offer the user a functional tablet experience are counted under the convertible NB category. Examples include Lenovo s Yoga Series and Dell s XPS 12 Ultrabook.

Detachable tablet

A detachable tablet meets all the criteria of a slate tablet but is designed and marketed to operate with a first-party keyboard designed specifically for the device. Further:

The keyboard component in detachable tablet devices can function as a simple keyboard and cover or include additional features such as ports, battery, HDD, and OS.

The keyboard component must be a first-party product designed specifically for the slate portion that creates a clamshell device through a physical connection (e.g., keyboard component size matches the slate portion and connects in a way that supports the screen as in a traditional clamshell device). Magnetic and/or resting

-

slate can happen over Bluetooth or another wireless frequency. The exception to the clamshell rule is if a device ships with a first-

s ENVY 8 Note.

The keyboard component may be optional or required. If the keyboard component is required, then the associated cost is included in the average value tracked; if it s

detachable tablet devices that have an optional keyboard component will count in the detachable tablet product category field, regardless of whether the keyboard

The detachable tablet category includes devices like Microsoft s Surface 3, Surface

s ENVY x2, Split x2, and SlateBook x2; and ASUS Transformer Book Trio.

roUS54462026 otes

The following relates to usage models or products that are not broken out in coverage:

Ruggedized systems are any notebooks designed and marketed as being ruggedized. Typically, these systems must meet MIL-STD-810F specifications, a set of standards designed by the U.S. government to test the durability of equipment under harsh conditions. These systems are tracked under the product traditional notebook.

Home servers are still an emerging category with very low volume. They fit into the desktop categories according to system volume.

MIDs are not considered PCDs and therefore are not included.

Technical egmentation

Screen ize band

This data tracks screen size by 1in. increments within the all- -one desktop, traditional notebook, ultraslim notebook, convertible notebook, mini notebook, detachable tablet, and slate tablet levels. Display size reflects the viewable image size measured as the distance between opposite corners (i.e., the diagonal). Coverage includes the following sizes:

0in. to <7in.

7in. to <8in.

8in. to <9in.

9in. to <10in.

10in. to <11in.

11in. to <12in.

12in. to <13in.

13in. to <14in.

14in. to <15in.

15in. to <16in.

16in. to <17in.

17in. to <18in.

18in. to <19in.

19in. to <20in.

20in. to <21in.

21in. to <22in.

22in. to <23in.

23in. to <24in.

24in. to <25in.

25in. to <26in.

26in. to <27in.

27in. to <28in

28in. to <29in.

29in.+

Touch

An optional component of IDC s Worldwide Quarterly Personal Computing Device Tracker covers touchscreen capability by all dimensions for historical periods and by region, country, product category, product, and consumer versus commercial for forecast periods.

Operating ystem

This represents the type of operating system being used from an end-user perspective for example, Windows, OS X, iOS, and Android. If a particular model could run more than one type of OS, we will track this as one model for

Note: OS data reflects initial configuration. We do not account for subsequent upgrades or

p rocessor i

Processor vendor. The vendor for the application processor is tracked at the OEM

recorded. IDC tracks the following list of vendors: Allwinner, Apple, Intel, MediaTek,

Processor speed band (detachable and slate tablet only).
measured in gigahertz (GHz) and fall into one of the following ranges:

- 0GHz <1.0GHz
- 1.0GHz to <1.2GHz
- 1.2GHz to <1.4GHz
- 1.4GHz to <1.6GHz
- 1.6GHz to <1.8GHz
- 1.8GHz to <2.0GHz
- 2.0GHz to <2.3GHz
- 2.3GHz+

Processor cores (detachable and slate tablet only). Some modern application processors have more than one CPU. IDC categorizes the number of actual CPUs in a handset into one of the following groups: single, dual, quad, hexa, and octa.

CPU type

s Worldwide Quarterly Personal Computing Device Tracker splits shipments by CPU type. Currently, this is limited to devices with ARM and x86 processors.

AI classification

This component of IDC s Worldwide Quarterly Personal Computing Device Tracker splits shipments based on a device s neural processing unit (NPU). IDC categorizes the devices NPU s capabilities measured in Tera Operations per Second (TOPS).

The following categories are available for desktops and notebooks:

- For Android, Chrome OS, Windows, and other OSs:
  - **Basic AI.** A device that offers 1 TOPS to <40 TOPS of NPU performance
  - **Gen AI.** A device that offers 40+ TOPS of NPU performance
  - **None.** A device without an NPU
- For macOS:
  - **Basic AI.** A device that offers 1 TOPS to <38 TOPS of NPU performance
  - **GenAI.** A device that offers 38+ TOPS of NPU performance
  - **None.** A device without an NPU

Note: Many PCs and workstations can run AI workloads through the use of a CPU or GPU; however, IDC is currently not tracking these details.

The following categories are available for detachable tablets and slate tablets:

- For Android, iOS, and other OSs:
  - **Basic AI.** A tablet that offers 1 TOPS to ≤30 TOPS of NPU performance
  - **Gen AI.** A tablet that offers >30 TOPS of NPU performance
  - **None.** A device without an NPU
- For Windows and Chrome OS:
  - **Basic AI.** A tablet that offers 1 TOPS to <40 TOPS of NPU performance
  - **Gen AI.** A tablet that offers 40+ TOPS of NPU performance
  - **None.** A device without an NPU

Note: IDC tracks AI data only from 1Q23. Hence all devices before 1Q23 will be marked as N/A.

Air interface/ eneration (detachable tablet and late tablet only)

IDC classifies each device with its highest possible air interface. For example, if a device is LTE ready and able to run on HSPA networks, the device is classified as an LTE device as that is the fastest or highest air interface. There is no direct correlation of the network technology that is available within a certain geography and the technology of the device that is shipped to that location. We have often found that vendors will ship devices that have a

t support its highest air interface strictly

3G air interfaces are supported such as CDMA versus HSPA, IDC will usually track the less globally common one where TD-SCDMA is less common than CDMA, which is less

The term *generation* is used to define a grouping of relative air interface technologies. A device can be a 2G, 2.5G, 3G, 4G, or a 5G device, depending upon the air interfaces it is

- 2.5G:
  - GPRS
  - EDGE
- 3G:
  - UMTS/WCDMA
  - HSPA
  - CDMA EV-DO
  - TD-SCDMA/TD-HSDPA
- 4G:
  - FD-LTE
  - TD-LTE
  - TD/FD-LTE
  - WiMAX
- 5G:
  - 5G New Radio (NR) forecast only

New air interfaces will be added as they become available.

Connectivity (otebook, detachable tablet, and late tablet only)

s Worldwide Quarterly Personal Computing Device Tracker splits shipments by the type of wireless connectivity offered: none, Wi-Fi, Wi-Fi/2G, Wi-Fi/3G, Wi-Fi/4G, and Wi-Fi/5G.

Integrated cellular (otebook, detachable tablet, and late tablet only)

s Worldwide Quarterly Personal Computing Device Tracker splits shipments by a device s ability to connect to a cellular network without any additional effort (see Table 4).

TABLE 4

PCD connectivity to integrated cellular

Connectivity	Integrated cellular
	No
	Yes
	Yes
	Yes

Source: IDC, 2025

RAM (detachable tablet and late tablet only)

s Worldwide Quarterly Personal Computing Device Tracker splits shipments by embedded SDRAM or DDR RAM capacity, which is tracked in gigabytes (GB).

Screen esolution detachable tablet and late tablet only)

s Worldwide Quarterly Personal Computing Device Tracker counts shipments by pixel

Storage (detachable tablet and late tablet only)

s Worldwide Quarterly Personal Computing Device Tracker splits shipments by

Voice calling (detachable tablet and late tablet only)

A device s ability to make voice calls over the cellular network out of the box without any

yes or no. Voice calling may

Customer egments

s user segmentation represents the final purchase of PCDs by each category of end users, whether those units are purchased directly from the vendor or indirectly via other channels. IDC counts the PCD toward the segment that paid for the device regardless of whether they are the actual end user. An example of this would be when a school purchases PCDs for education and students are the final end users, the PCD shipment would be counted under education and not consumer because the school paid for the device. With regard to business segments, any multinational company is recorded as its global employee size (e.g., IDC still counts a multinational company that has 1,000+ employees in the United States but only 20 employees in Poland as a very large business).

Segments include:

Consumer. All home purchases, regardless of usage (home office, work at home,

Small office. Nonresidential businesses with fewer than 10 employees in the country

Small business. Businesses with 10 99 employees in the country

Medium-sized business. Businesses with 100 499 employees in the country

Large business. Businesses with 500 999 employees in the country

Very large business. Businesses with 1,000+ employees in the country

Government. City, county, state, provincial, regional, military, and federal -owned commercial enterprises.)

Education. Education and instructional institutions covering K consumer segment. Education includes both public and private institutions. Shipments to public

Customer egment roups and otes

Consumer. The home category is formally referred to as the consumer market.

Public. The sum of the government and education segments is sometimes referred to collectively as the public sector.

Business. The sum of small office, small business, medium-sized business, large business, and very large business is sometimes referred to collectively as the business sector.

Commercial. The sum of public and business (all non consumer shipments) is sometimes referred to collectively as the commercial sector.

Required student devices:

In cases where a school buys a device for a student (e.g., required purchase, included in tuition), the institution still makes the product selection and purchase, so the shipment is counted under education.

In cases where a school requires a student to bring a device (purchase if necessary), but the student pays directly to the vendor and can choose the specific product, the shipment is counted under consumer reflecting the student s role in making the purchase.

Channel/channel roups

IDC categorizes channels at two levels, channel group and channel. The channel group represents an aggregation of channel data in two categories, direct and indirect. Direct is anything sold by the vendor directly to the end user, while indirect captures purchases that first pass through a third party.

The channel data in IDC trackers represents the channel from which end users purchase new devices. IDC does not track intermediate movements such as vendor sales to first-distributors. For example, if a vendor sells to a mobile operator that then sells on to a retailer, IDC will capture the channel as retail, not telco. Channels are defined based on their primary business activity. Channel classification of companies that are subsidiaries of other companies (e.g., CDW Logistics is a subsidiary of CDW) will be evaluated independently. Within each company, IDC aims to capture all sales to end users within a single category. This will apply to in-house operations that are a minority of sales such as online sales by retailers as well as retail or value-added reseller (VAR)-type operations of online companies. However, when sales are not to end users (e.g., they can be to other channels), those shipments will be captured where the end user buys from (e.g., the last level of channel).

s channel methodology tracks how the end user places the order, not how the order is fulfilled or received. For example:

If a customer walks into a retail store and buys a PC, but the PC is shipped to the customer, IDC counts that as retail and not direct. Conversely, if a user orders a MacBook online but walks into an Apple store to pick it up, that would be internet and not store.

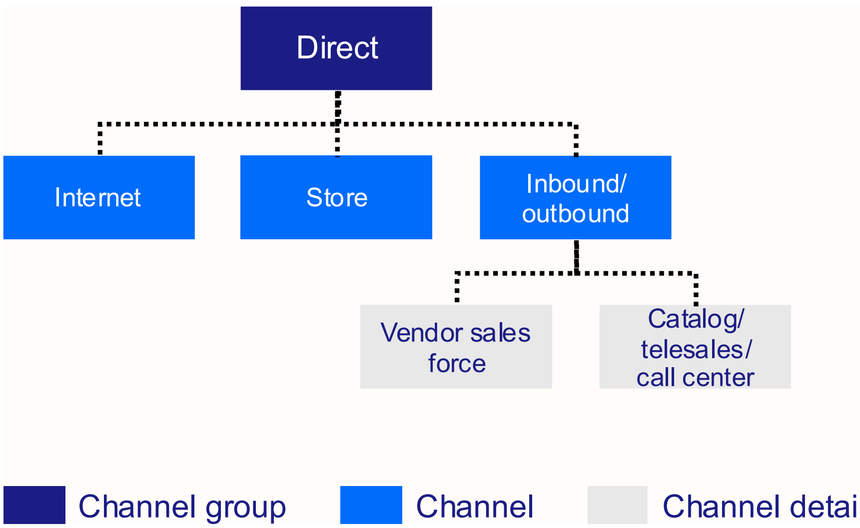
If a global headquarters makes a sale that is delivered locally by partners, this would count as a direct sale. Effectively, the delivery and potential installation and support services could be considered add-ons or outsourced parts of a deal, but the sale of the primary hardware would count as direct, even if a local partner is compensated for the sale in its sales territory. Conversely, if a deal is signed by a local channel, but delivery or related services are provided directly by the vendor, we would count the sale under the channel partner. For enterprise sales in which multiple parties contribute, IDC will attempt to classify the sales channel based on where the contract was signed and who invoices the client.

A channel does not have to be officially approved by a vendor. In some cases, a channel that is not authorized by or directly involved with a vendor will nevertheless offer the vendor s products. This can include selling products that are not officially targeted for or launched in a specific country but are nevertheless offered by a channel sourcing products from a nonvendor source (e.g., other channel).

Figure 2 details the channels captured in the direct channel.

FIGURE 2

s channel t d



Source: IDC s Worldwide Quarterly Personal Computing Device Tracker, 2015

Note: Channel detail is not broken out in IDC s tracker products.

internet

An internet sale includes sales made directly by the vendor to the end users through the vendor s website or its online stores. Purchase confirmation must be made via the internet. Internet telesales, in which a product is purchased online but some human element is involved in the process (e.g., a customer calls the vendor directly to talk before or transaction), will be included in this channel.

Sometimes vendors will outsource the operation of their website. Most cases where a third party manages invoicing and fulfillment would count as an indirect sale (generally, etailer). -branded sites (e.g., Lenovo.com, HP.com), the vendor should own the brand and domain and have influence over site design and other aspects so that, even if operational logistics such as invoicing and fulfillment are handled by a third party, sales through these vendor-branded websites will count as direct internet.

Exceptions: Sales made via dealer or retailer websites will not be included here. For s website redirects to another website (e.g., that of a dealer or a retailer) during the sales transaction, then the shipment is counted within the dealer/VAR/systems integrator (SI) or retail channel itself. Similarly, online sales by retailers or dealers that are generated independently from a vendor s website or online store will be ot as an internet sale under the direct channel group.

The store channel includes sales made by the vendor to the end user through storefront businesses that are owned and supplied directly by the vendor. These stores typically sell to a large number of unrelated customers, as is the case in the retail channel, with customers initiating the sales transactions. Examples include sales made through Apple stores or

i o

Inbound/outbound includes all vendor direct sales that do not occur over the internet or in vendor-owned stores. The two channel detail breakouts in inbound/outbound are:

- **Vendor sales force (i.e., outbound):** These are sales made directly by the vendor to the end user through its sales force, agents, or representatives (who outwardly

  accounts. Examples include sales generated by the direct sales force of international

- **Catalog/telesales/call center:**
  end user via catalog, telesales, or call centers whereby an end user calls or otherwise connects inbound to the vendor.

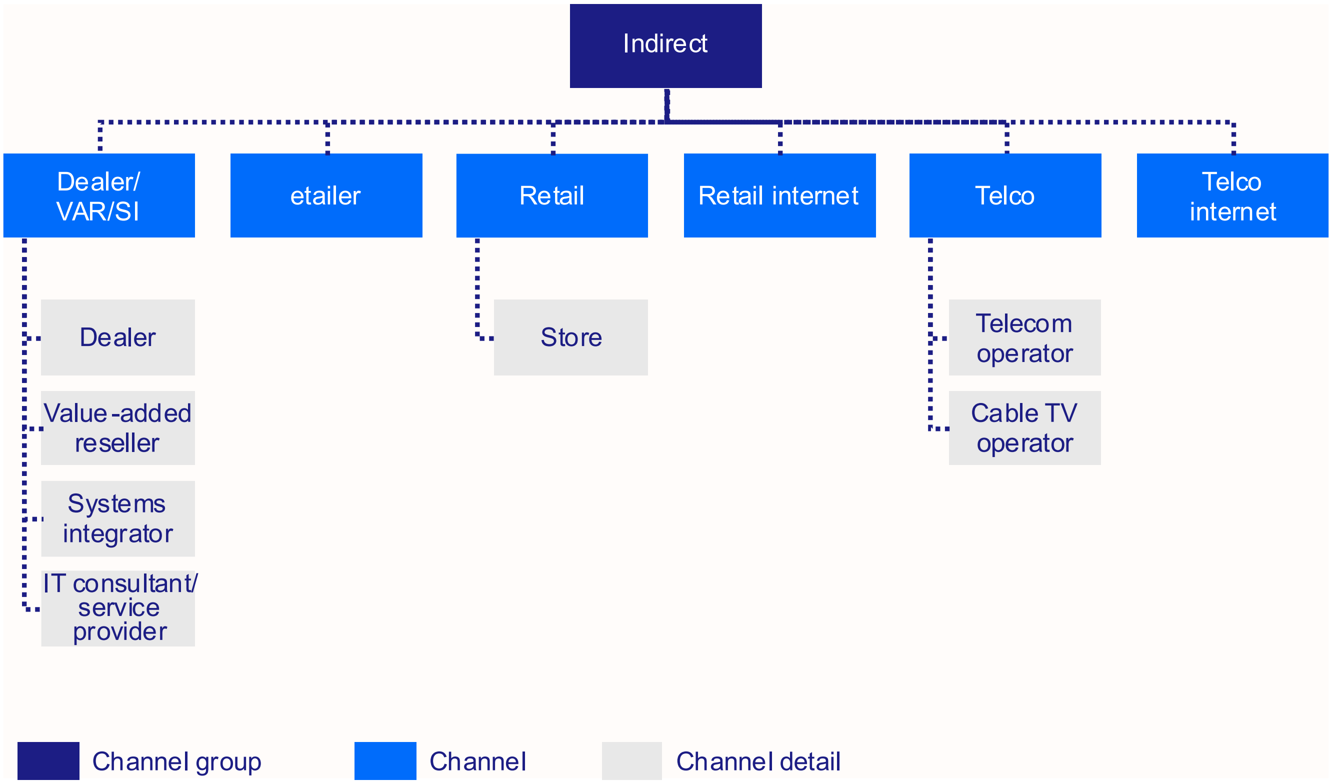
In cases in which a gift is offered with a purchase (gift with purchase), for example, where a tablet is offered as a giveaway with a newspaper subscription, IDC would track this as inbound/outbound because the newspaper company does not represent a retail outlet from which a customer could buy the device.

Exceptions: Sales in which the vendor plays a significant role in developing the business but does not directly invoice or sign a sales contract with the end user are not considered inbound/outbound sales and are instead tracked under the indirect channel group.

Figure 3 details the channels captured in the indirect channel.

FIGURE 3

s channel t indirect



Note: Channel detail is not broken out in IDC s tracker products.

Source: IDC s Worldwide Quarterly Personal Computing Device Tracker, 2015

Indirect dealer/VAR/SI

The dealer/VAR/SI channel is characterized by firms that offer value by bundling extra services with the product sale beyond both price and availability (which are generally the primary competitive points of retail outlets). Only sales to the end user (i.e whether a company or consumer) would qualify in this channel. For example, if CDW acts

counted under retail while those sold to enterprises

are large channel partners that have national coverage and - -sized and large businesses. Each with an annual turnover of more than $10 million, of which more than 50% is made through large accounts, and they will usually represent t
products, including but not limited to PCs, tablets, phones, entry-

midrange systems, networks, peripherals, and software. Their business models tend to involve a greater share of services than those of smaller dealers. Examples include

Traditional dealers are IT-dedicated resellers (predominantly IT product resellers) that sell the same standard of products and services as corporate dealers. They generally do more than 60% of their business on the resale of third-party products but less than 20% from internally created services. Traditional dealers/resellers address a large range of customers, from consumers to medium-sized businesses, depending on the size, product portfolio, customer basis, and geographical situation of the customers (from small shops to larger businesses addressing professional users and)

Value-added resellers typically derive a significant portion of their revenue from services in addition to product resale. Examples of VARs include FusionStorm and

- -related revenue from systems integration, implementation, configuration, custom application development, and/or consulting services. IDC defines SIs as organizations that conduct the planning, design,
of a solution that addresses a customer s specific technical or business needs. SIs
s decision to purchase a particular product; they may
or influence fee from the vendor or resell the vendor
Examples include Accenture, Infosys, Capgemini, and Deloitte.

that area. Others have expertise in a broader set of technologies but focus on a specific geographic market.

Service providers are firms that earn the majority of their services-related revenue from providing hosted or cloud-based services on an ongoing basis. Service providers normally have ongoing customer relationships as opposed to time-limited project-based work. The service provider category includes managed service providers/outsourcers, telecom service providers, hosting providers, cloud service providers, business process service providers, and authorized third-party maintenance companies.

Distributors typically provide warehousing, order processing, credit, and shipping services. Distributors direct sales to end users are tracked in the dealer/VAR/SI channel.

Exceptions:

Shipments made to any non-final-tier entity such as distributors or resellers are not considered. These sales would be captured at the point of sale from the final tier to the end user.

If the dealer/VAR/SI is acting solely as a fulfilment agent on behalf of a vendor, where the vendor has negotiated the terms of the deal and manages the invoicing, the units would be counted under direct.

Table 5 provides examples of dealers/VARs/SIs by region.

TABLE 5

Examples of dealers/VARs/SIs by

United States	Canada	Japan	Europe, the Middle East, and Africa (EMEA)	Asia/Pacific	Latin America
Avnet	OnX	Otsuka Shokai	Computacenter, Bechtle, Insight, Systemax, and Capgemini (WE)	Roxy Mas and Metrodata (Indonesia)	Brightstar
CopyFax, Insight Direct USA, SHI International, Zones, and SoftChoice Corp.	Compugen	Too Corp.	Redington, Metra Computer, FDC International, Achievers, Musallam Trading, ABC Data, Action and AB IT, ASBIS, Incom, Komputronik, Connect Distribution, Westcomdist, System Tech, Computer Warehouse Group, Allied Computers, Telnet, CBC, Equipment Hall, TECHNOTRADE, Treolan, MICS, Resource Media, Marvel, OCS, Staten, and Alpha Data	BrightPoint and Brightstar (New Zealand)	Ação Informática, AGORA Telecom, Officer, Simpress, and CTIS (Brazil)
CDW Logistics	Metafore		Arti&Arti, KoçSistem, Sentim, Servus, Netas, and Mas Deha (MEA)	BBS Distributions and SNS Network (Malaysia)	Adexus and Sonda (Chile)
Cognizant	Mégaburo		Interlink, Asteros, Avtonim, Linko, and NBZ Computers (CEE)	Synnex (Australia)	Solution Box and Synapsis (Argentina)
Teknologize	RGO Office Products			Kingdee and Beijing Weifeng Fuda Technology Co. Ltd. (PRC)	SyC, Grupo Scanda, and Migesa (Mexico)
Infosys	Buropro			Integrated Computer System (ICS) (Philippines)	ALKOSTO and Exito (Colombia)

TABLE 5

Examples of dealers/VARs/SIs by

United States	Canada	Japan	Europe, the Middle East, and Africa (EMEA)	Asia/Pacific	Latin America
Westcon	Nova Networks Inc.			JOS (Hong Kong)	
Insight Enterprises	Graycon Group Ltd.				

Note: This list is not exhaustive, and the companies listed may operate in multiple regions. The examples are based on the current operations of these companies, which may have changed from earlier periods such as their original focus.

Source: IDC s Worldwide Quarterly Personal Computing Device Tracker, 2025

Indirect tailer

This category includes both pure etailer (web based) and other low-overhead channels, such as third-party mail, that exclusively or predominantly sell through catalogs and direct mail. This includes online marketplaces such as Amazon.com and eBay, which facilitate online sales from third parties, ranging from individuals to resellers to vendors themselves. Business-- -business (B2B) office suppliers, print and electronic catalogs, and TV sales (e.g., home shopping networks) are included in this category. IDC tracker shipments only ts of new products and do not include the used product sales commonly found in etailers.

In some cases, such as when an etailer branches into bricks-and-mortar stores, IDC

conducted online.

In countries such as China, where online stores such as Tmall.com host subsites that are like a virtual store within a store, whether for a vendor or retailer, but form a part of the larger Tmall website, we would count these as etailer units.

Exceptions:

- Internet sales from vendors (via their own websites and online stores) are tracked internet (e.g., Apple.com and Dell.com).
- Internet sales by indirect channels are tracked under their respective channel categories (e.g., BestBuy.com is tracked under the retail internet).
- Products branded with an etailer internet channel, not the etailer channel. For example, the Amazon Fire phone and tablet sold on Amazon.com would be counted as direct internet rather than etailer, since Amazon is the vendor.

Table 6 provides examples of etailers by region.

TABLE 6

e tailers by

United States	Canada	Japan	EMEA	Asia/Pacific	Latin America
Amazon	Newegg	Rakuten	Cdiscount and Pixmania (WE)	JD, Tmall, and Suning Yigou (PRC)	Submarino, TIM, Vivo, and Movistar (Brazil)
Dabs	Expansys	Amazon.co.jp	Cyberport (Germany)	Flipkart and Snapdeal (India)	Kabum (Brazil)
eBay	NCIX.com	Yahoo!	Ebuyer and Argos (United Kingdom)	Gmarket and Digital Plaza (Korea)	Linio and Telmex (Mexico)
Wirefly, GovConnection, TigerDirect, PC Connection, and PCM Sales	DirectCanada	Japanet Takata	Souk, Jumia, JadoPado, hepsiburada.com, GittiGidiyor.com, n11.com, akakce.com, and Cimri.com (MEA)	Kogan (Australia)	MercadoLibre
		ASKUL	Awok, Konga, Dealdey, torx.pl, RAM.net.pl, DeniKomp, CSD, NISSA, PIRIT, ABIUS, and Copydom (CEE)	PChome (Taiwan)	OLX
				Qoo10 and Lazada (Southeast Asia)	alaMaula
					Privalia

Source: IDC s Worldwide Quarterly Personal Computing Device Tracker, 2025

Indirect tail

The retail channel is primarily characterized by storefront businesses that sell to a large number of unrelated customers, generally consumer and small-office users. In the retail channel, sales are generally initiated by the customer and a large percentage of sales are transactional, meaning they are the result of an informal and immediate purchase of goods and/or services. Sales are typically generated from walk-in business rather than from established relationships or contracts or because of deliveries/installations on clients premises (though some retail sales may still offer small-scale delivery and installation). Retail outlets typically compete on price and availability and depend less on services than

A large portion of a retail location is dedicated to product display, in contrast to dealers, which may have a showroom but sell more from price lists than from product displays. Other

PC and IT malls: In many regions (particularly emerging markets), small independent shops are often colocated in PC/IT malls. While individual shops may be limited in scope and dedicated to only a few brands and/or products, the combined location acts as a brand-- transactional outlet. As such, sales through these malls are counted under retail.

Open-air stalls and kiosks: Open-air markets and stalls (often found in emerging markets), many of which remove stock for overnight storage elsewhere, are tracked under the retail channel. Kiosks that are found within a larger singly owned retail environment in which any portion of the sales revenue goes to the owner of the retail environment are tracked under the retail channel.

Exceptions: Operator-branded kiosks that keep 100% of sales revenue are tracked under the telco channel.

Franchise outlets: Outlets dedicated to (and branded with) a vendor name (e.g., Sony outlets) fall under the retail channel, not under direct store, since franchises are generally not vendor owned. Examples of retail include mass merchants (including mass merchandisers, wa
showrooms, and toy stores), consumer electronics retailers, office product

high street

store
and not retail. For example, Tesco tablets sold via its own storefronts would be under
-and-mortar retailers such as
BestBuy.com a

TABLE 7

Examples of

United States	Canada	Japan	EMEA	Asia/Pacific	Latin America
EDION	Glentel	Bic Camera	Auchan, Carrefour, and Dixon Group (WE)	Dixintong, Suning, GOME, and Guangzhou Tianhe (PRC)	Costco
Best Buy	EDION		Carphone Warehouse and Currys (United Kingdom)	The Mobile Store and Croma (India)	Casas Bahia, Extra, Leader, Magazine Luiza, B2W, Fnac, Kalunga, and Nagem (Brazil)
Walmart	Canada Computers	K	Emax, Lulu Supermarkets, Tesco, PC Outlet, Pack Shop, and Mediamark (CEMA)	Erafone and OkeShop (Indonesia)	Falabella, PC Factory, Paris, and La Polar (Chile)
Staples	The Source	Yodobashi	Euroset, Svyaznoi, M.Video, Eldorado, Bely Veter, Media Markt, Technosila, PhotoPlus, and DNS (Russia)	Jaymart (Thailand)	
TK3C (Taiwan)	Frávega, Falabella, and Garbarino (Argentina)				
	Staples	Yamada	Axiom (GCC)	Harvey Norman and Dick Smith (Australia)	Office Depot, OfficeMax, Coppel, Elektra, El Palacio de Hierro, and Liverpool (Mexico)
			Teknosa, Media Markt, Bimeks, Vatan, and Gold (Turkey)	JB Hi-Fi, The Warehouse, and Noel Leeming (New Zealand)	Efe (Peru)
				Hi-Mart, Homeplus, and Emart (Korea)	Siragon (Venezuela)
				Broadway and Fortress (Hong Kong)	
				Tsann Kuen (Taiwan)	
				Challenger (Singapore)	

Note: This list is not exhaustive, and the companies listed may operate in multiple regions. The examples are based on the current operations of these companies, which may have changed from earlier periods such as their original focus.

Source: IDC s Worldwide Quarterly Personal Computing Device Tracker, 2025

Indirect telco

The telco channel includes sales of devices by an entity that is licensed to operate cellular services. These devices are usually meant to function on the operator s network, though

channel from which the end user made the purchase from. For example, an end user could buy an unlocked product from an operator and then swap the
or choose to only use Wi-Fi for connectivity.

IDC tracks devices under telco through own-

operators of large-

such as cable TV operators that also hold cellular licenses (e.g., StarHub) or mobile virtual network operators (MVNOs) that have entered into agreements wit

Telco sales include products sold alone and those bundled with voice/data/internet subscriptions and do not represent the total number of products bundled with voice/data/internet subscriptions (prepaid or postpaid), as those bundles are also available through other channels such as retail and etailer. The average value of a product sold with a subsidy from a telecom provider is tracked as the retail value of the product without the subsidy.

Exceptions:

- In the case of a bundled offering between a vendor and a telecom provider that is . This applies to cases such as a Verizon Wireless data plan purchased along with a tablet from Best Buy and an AT&T voice plan purchased along with a mobile phone from Walmart.
- Products branded under a telco are counted under Indirect Telco as multiple brands are carried at the outlet. For example, a Vodafone tablet sold via a Vodafone Telco.
- Specialized telco stores, which offer mobile voice/data plans from multiple operators and have multivendor device portfolios, are tracked under the retail channel. In recent years, some major specialized telco stores have either exited the market (e.g., Phones4U) or have merged with retailers (e.g., Carphone Warehouse acquired by Dixon).
- In a franchise scenario, if more than one telco s services are offered, these units are no longer tied to the telco and are instead counted under retail.
- In cases where an operator has ownership of a retailer under a different brand name, we would count the channel at the most granular level, which in this case is retail. For

example, Telus in Canada owns the retailer Blacks, and we would count those under retail.

Table 8 provides examples of telcos by region.

TABLE 8

Examples of telcos by

United States	Canada	EMEA	Asia/Pacific	Latin America
	Fido	Etisalat and Zain (GCC)	Celcom, Maxis, DiGi, U Mobile, and Yes (Malaysia)	Entel
		Airtel	Viettel, MobiFone, and Vinaphone (Vietnam)	Tigo
		MTN and Vodacom (Africa)	Telstra, Optus, Vodafone, Skinny, 2degrees, and Spark (Australia and New Zealand)	Digicel
		Globacom	CSL, CMHK, 3, and SmarTone (Hong Kong)	Telcel
		MTS, Beeline, and MegaFon (Russia/CIS)	Smart and Globe (Philippines); SK Telecom, Olleh, and LG Uplus (Korea)	Iusacell
			SingTel, StarHub, and M1 (Singapore)	Movilnet

Note: This list is not exhaustive, and the companies listed may operate in multiple regions. The examples are based on the current operations of these companies, which may have changed from earlier periods such as their original focus.

Source: IDC s Worldwide Quarterly Personal Computing Device Tracker, 2025

Indirect etail internet

The retail internet channel includes online sales made by the retailers to the end users .

Table 9 provides examples of retail internet by region.

TABLE 9

Examples of etail internet by

United States	Canada	Japan	EMEA	Asia/Pacific	Latin America
	Bestbuy.ca, Walmart.ca, Staples.ca	Yodobashi.com, yamada-denkiweb.com, biccamera.com	www.fnac.com, www.mediamarkt.de/	www.harveynorman.com.au, www.mxmemoxpress.com,	www.casasbahia.com.br, Liverpool.com.mx, Fravega.com
			www.mvideo.ru	www.sundan.com, www.suning.com	

Note: This list is not exhaustive, and the companies listed may operate in multiple regions. The examples are based on the current operations of these companies, which may have changed from earlier periods such as their original focus.

Source: IDC s Worldwide Quarterly Personal Computing Device Tracker, 2025

Indirect   telco internet

The telco internet channel includes online sales made by the telecom providers other than its own devices to the end users through its website or its online stores (e.g., all devices Telstra   -branded phones that are sold on telstra.com will be counte internet channel).

Table 10 provides examples of telco internet by region.

TABLE 10

Examples of telco internet by

United States	Canada	Japan	EMEA	Latin America
	bell.ca, telus.com, rogers.com	www.softbank.jp/online-shop, www.nttdocomo.co.jp	www.shop.ee.co.uk, www.boutique.orange.fr	Entel.cl, Telcel.com, www.movistar.com.ar
			moscow.megafon.ru, www.play.pl	

Note: This list is not exhaustive, and the companies listed may operate in multiple regions. The examples are based on the current operations of these companies, which may have changed from earlier periods such as their original focus.

Source: IDC s Worldwide Quarterly Personal Computing Device Tracker, 2025

Geographic egmentation

IDC segments the worldwide market by the following geographic regions:

- United States
- Canada
- Latin America
- Western Europe
- Central and Eastern Europe (CEE)
- Middle East and Africa (MEA)
- Asia/Pacific (excluding Japan and China) (APEJC)
- Japan
- PRC

Table 11 shows IDC s PCD market segmentation by geography. Table 12 indicates where territories and countries not specifically named in the tracker are counted.

TABLE 11

s PCD market egmentation by eography

States	Canada	Japan	PRC	Asia/Pacific (ex. Japan and China)	Western Europe	Central and Eastern Europe	East and Africa	Latin America
	Canada	Japan	PRC	Australia	Austria	Bulgaria	Algeria	Argentina
				Bangladesh	Belgium	Croatia	Bahrain	Bolivia
				Hong Kong	Denmark	Czech Republic	Botswana	Brazil
				India	Finland	Estonia	Egypt	Chile
				Indonesia	France	Hungary	Ethiopia	Colombia
				Korea (South)	Germany	Kazakhstan	Ghana	Costa Rica
				Malaysia	Greece	Latvia	Israel	Dominican Republic
				Myanmar				
				New Zealand	Ireland	Lithuania	Jordan	Ecuador
				Philippines	Italy	Poland	Kenya	El Salvador
				Singapore	Netherlands	Romania	Kuwait	Guatemal a
				Sri Lanka	Norway	Russia	Lebanon	Mexico
				Taiwan	Portugal		Morocco	Panama
				Thailand	Spain	Slovakia	Namibia	Paraguay
				Vietnam	Sweden	Slovenia	Nigeria	Peru
				Asia/Pacific	Switzerland	Ukraine	Oman	Puerto Rico
					Kingdom	Central and Eastern Europe	Pakistan	Central America

TABLE 11

s PCD market egmentation by eography

States	Canada	Japan	PRC	Asia/Pacific (ex. Japan and China)	Western Europe	Central and Eastern Europe	East and Africa	Latin America
							Qatar	Latin America
							Saudi Arabia	Uruguay
							South Africa	Venezuela
							Tanzania	
							Tunisia	
							Turkey	
							Uganda	
							Rest of	
							East	
							Africa	

Note: Additional countries in CEE, MEA, Latin America, and Asia/Pacific are tracked or analyzed on a biannual or an annual basis.

Source: IDC s Worldwide Quarterly Personal Computing Device Tracker, 2025

TABLE 12

s PCD market country t

Region	IDC country (quarterly data available)	Country
	United States	United States
	Canada	Canada
	Japan	Japan
	PRC	PRC
	Austria	Austria
	Belgium	Belgium and Luxembourg (Belgium)
	Denmark	Denmark, Faroe Islands (Denmark), Greenland (Denmark), and Iceland (Denmark)
	Finland	Finland
	France	France, Monaco, and Saint Pierre and Miquelon (France)
	Germany	Germany
	Greece	Greece, Cyprus
	Ireland	Ireland
	Italy	Italy, Malta, San Marino, and Vatican City Holy See
	Netherlands	Netherlands
	Norway	Norway, Svalbard
	Portugal	Portugal
	Spain	Spain, Andorra (Spain), Canary Islands (Spain), and Gibraltar
	Sweden	Sweden
	Switzerland	Switzerland, Liechtenstein
	United Kingdom	United Kingdom, Guernsey, Isle of Man (United Kingdom), and Jersey

TABLE 12

s PCD market country t

Region	IDC country (quarterly data available)	Country
Central and Eastern	Bulgaria	Bulgaria
	Croatia	Croatia
	Czech Republic	Czech Republic
	Hungary	Hungary
	Poland	Poland
	Romania	Romania
	Russia	Russia
	Slovakia	Slovakia
	Slovenia	Slovenia
	Ukraine	Ukraine
	Latvia	Latvia
	Lithuania	Lithuania
	Estonia	Estonia
	Kazakhstan	Kazakhstan
	Rest of Central and Eastern Europe	Albania, Armenia, Azerbaijan, Belarus, Bosnia and Herzegovina, Franz Josef Islands (Russia), Georgia, Kosovo, Kyrgyzstan, Macedonia, Moldova, Montenegro, Tajikistan, Turkmenistan, and Uzbekistan
	Egypt	Egypt
	Israel	Israel
	Saudi Arabia	Saudi Arabia
	South Africa	South Africa

TABLE 12

s PCD market country t

Region	IDC country (quarterly data available)	Country
	Turkey	Turkey
	United Arab Emirates	United Arab Emirates
	Bahrain	Bahrain
	Kuwait	Kuwait
	Oman	Oman
	Qatar	Qatar
	Jordan	Jordan
	Lebanon	Lebanon
	Algeria	Algeria
	Botswana	Botswana
	Ethiopia	Ethiopia
	Ghana	Ghana
	Kenya	Kenya
	Morocco	Morocco
	Namibia	Namibia
	Nigeria	Nigeria
	Tunisia	Tunisia
	Tanzania	Tanzania
	Uganda	Uganda
	Rest of Middle East	Afghanistan, Iran, Iraq, Syria, West Bank and Gaza, and Yemen

TABLE 12

s PCD market country t

Region	IDC country (quarterly data available)	Country
	Rest of Africa	Angola, Benin, Burkina Faso, Burundi, Cameroon, Cape Verde, Central African Republic, Chad, Comoros, Congo, Democratic Republic of the Congo, Republic of the Cote d Ivoire, Djibouti, Equatorial Guinea, Eritrea, Gabon, Gambia, Guinea, Guinea-Bissau, Lesotho, Liberia, Libya, Madagascar, Malawi, Mali, Mauritania, Mauritius, Mayotte (France), Mozambique, Niger, Reunion (France), Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, Somalia, Sudan, Swaziland, Togo, $VLD3DFLILF
(excluding Japan and China) China)	Australia	Australia
	Bangladesh	Bangladesh
	Hong Kong	Hong Kong (China)
	India	India
	Indonesia	Indonesia
	Malaysia	Malaysia
	New Zealand	New Zealand
	Philippines	Philippines
	Singapore	Singapore
	South Korea	South Korea
	Sri Lanka	Sri Lanka
	Taiwan	Taiwan
	Thailand	Thailand
	Vietnam	Vietnam

TABLE 12

s PCD market country t

Region	IDC country (quarterly data available)	Country
	Rest of Asia/Pacific	American Samoa (United States), Bhutan (India), Brunei, Cambodia, Christmas Island (Australia), Cook Islands (New Zealand), East Timor, Fiji, French Polynesia (France), Guam (United States), Keeling Islands (Australia), Kiribati, Laos, Macau (PRC), Maldives, Marshall Islands, Micronesia, Mongolia, Myanmar (formerly Burma), Nepal, New Caledonia (France), Niue (New Zealand), Norfolk Island (Australia), North Korea, Northern Mariana Islands (United States), Palau, Papua New Guinea, Pitcairn Islands (United Kingdom), Samoa, Solomon Islands, Tokelau (New Zealand), Tonga, Tuvalu, Vanuatu, Wallis and Futuna (France), and Nauru (Australia)
	Argentina	Argentina
Latin America	Bolivia	Bolivia
	Brazil	Brazil
	Chile	Chile
	Colombia	Colombia
	Costa Rica	Costa Rica
	Dominican Republic	Dominican Republic
	Ecuador	Ecuador
	El Salvador	El Salvador
	Guatemala	Guatemala
	Mexico	Mexico
	Panama	Panama
	Paraguay	Paraguay
	Peru	Peru
	Puerto Rico	Puerto Rico
	Uruguay	Uruguay
	Venezuela	Venezuela

TABLE 12

s PCD market country t

Region	IDC country (quarterly data available)	Country
	Rest of Central America	Belize, Honduras, and Nicaragua
	Rest of Latin America	Anguilla (United Kingdom), Antigua and Barbuda, Aruba (Netherlands), Bahamas, Barbados, Bermuda (United Kingdom), Cayman Islands (United Kingdom), Cuba, Dominica, French Guiana (France), Grenada, Guadeloupe (France), Guyana, Haiti, Jamaica, Martinique (France), Montserrat (United Kingdom), Netherlands Antilles (Netherlands), Saint Helena (United Kingdom), Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, South Georgia Island (United Kingdom), Suriname, Trinidad and Tobago, Turks and Caicos Islands (United Kingdom), Virgin Islands (United Kingdom), and Virgin Islands (United States)

Source: IDC s Quarterly PCD Tracker, 2025

m m

Company, vendor, and brand ecognition of ownership of PC hipments

reflects, in the current IDC reporting period, the financial, legal, or corporate group entity that exists as the outcome of mergers or acquisitions as if it

periods. This data field simplifies trend analysis for companies that have merged

Vendor reflects, in the current reporting period, the financial, legal, or corporate

reflects the vendor brand name (logo) on the box. For example, Alienware is the brand name as it appears on the box, whereas the Dell Inc. is the parent company.

Operating vendor and date of acquisition

IDC defines an operating vendor based on ownership of greater than 50% interest in a company, and the parent company must be a vendor that continues to actively produce and sell the acquired products and services. This definition explicitly excludes financial holding companies, private equity firms, and so forth as vendors in IDC trackers. When companies merge or are acquired, IDC tracker data will represent the new entity starting in the research cycle in which the deal is legally confirmed.

Shipments

Unit shipments are a measure of the number of new PCDs shipped by a vendor to all distribution channels or directly to end users. Units are counted as the title is transferred from the vendor to a channel or customer, provided this is consistent with other checks. This aligns with supply-side vendor financial information and also means that shipments may be counted in their destination country before they physically arrive (provided the title has been transferred). We continue to count on a unit basis, not as revenue is received or billed (which is particularly necessary for purchases on credit). For example, if a vendor sells product in Miami to a channel in Latin America, the title may be transferred as part of a sales contract signed in Miami while the physical product takes time to ship and pass customs before being physically received by the channel in the country. In this case, the shipment would be counted *when* the title is transferred (in Miami, aligning with the vendor s sales contract). The shipment would still be counted *where* it ends up (in Latin

Shipment data are validated using a variety of sources, which can include information about device assembly, component supply, import records, reexports, large projects, inventory, and shipments to or from specific channels. Some differences in these sources may exist because of timing or partial market coverage. To address this, IDC considers which sources are most reliable and evaluates the alignment over time. As a result, adjustments to align multiple sources may not happen every quarter but are applied when discrepancies persist

s worth noting that the

which is different than timing factors that can affect specific quarters but balance out over time. PCDs are not double counted in
product may be manufactured by an OEM, then shipped to a vendor, and then shipped to a channel or end user). In this

New PCDs are counted in our PCD tracker, whether sold complete through a recognized vendor or channel or built by individuals or micro-assemblers that purchase loose
- -
Sales of used PCDs and upgrades of used PCDs are not counted in IDC s Worldwide

Related to upgrades versus rebuilt PCDs, IDC does not count upgrades in PCD shipments. This applies when one or more components are replaced to fix a specific issue on a used PCD (such as adding more storage or memory). If a PCD is essentially rebuilt (replacing motherboard, processor, HDD, memory, and so forth generally the valuable core components simultaneously but not necessarily replacing the keyboard, mouse, or case generally the low-value components), then IDC considers this a new PCD. In these -base purposes and also eliminates the potential sale of an assembled PCD. We believe that rebuilt PCDs represent a small part of the market and are generally limited to individuals building their own systems or small local assemblers serving small local markets.

If a shipment is held in inventory by a local subsidiary, it is not counted until it is sent into the channel. Local subsidiaries are not considered part of the channel.

If a product is returned to a vendor, for example through some type of price protection agreement with channels, then those returns should be deducted from the count of new shipments. If time has elapsed so that a shipment was counted in one quarter, but t return happened in the following quarter, then the deduction should happen when the return occurs. This same approach can be used for products returned by users within a short warrantee period if the product can be resold as new. If the product can t be resold as new, then the original shipment stands, and no return needs to be deducted. Any refurbishment and resale of such a product would be considered a used system, and resale would not qualify as a new shipment in IDC s trackers.

In OEM situations, shipments are attributed to the company whose name appears on the final product. In some countries, it is common for the OEM s name to appear on the back of a product for units sold through large government deals or sourced for large companies. In these cases, if the manufacturer name is clearly stated on the back of the product, IDC will consider the manufacturer name. This is most common for PC monitors in Brazil.

In cases where a significant amount of products are sold through a channel in one country but are actually reexported and used in another, they are counted only in the country of final use (e.g., products sold into Dubai [UAE] en route to other MEA countri applies to shipments reexported from Dubai (UAE), in some cases within the CIS, from Miami to Latin America, from Hong Kong to other markets, and a few other markets. IDC does not attempt to reflect the movement of products by individuals at an individual level but does attempt to reflect the commercial redistribution of products in volume.

New products consumed by the company that manufactures them should be counted as shipments (e.g., even if they are not technically shipped to a third party) as they reflect total demand, production, installed base, and so forth. Similarly, new products tha donated or given away for free should be counted as shipments. This reflects the fact that

provides PCDs free but sells a solution and maintenance, or a vendor gives channel partners free products based on achieving certain shipment volumes). Even a completely philanthropic gift contributes to satisfying demand, effectively competes with the alternatives (buying products), and adds to the installed base. Counting these systems also contrib to reconciling with component supplies and properly tracking the installed base of systems.

Grey market

IDC does not differentiate grey market shipments from others but tracks all shipments based
grey market includes everything that is
not 100% official up to and including anything that is illegal. This includes:

Use of distribution channels that are not officially supported by a vendor (This can

distribution channel behavior, and distributing to unapproved clients/countries.)

Official channels purchasing and reselling to partners that either run a carousel

Full or partial evasion of import, sales, or other taxes on complete systems as well as

Use of discounted or even secondhand components in the production of new

Sales of or supported by pirate or inappropriately licensed software or content

Sales in, ales through, and ales out

To provide the most accurate picture of the PCD market, vendor performance, and user

perform such analysis, IDC s Worldwide Quarterly Personal Computing Device Tracker methodology combines a mixed approach of sales in, sales through, and sales out projection. In detail:

Unit shipments are counted, as defined previously, once the vendor passes ownership to a third party (either end user or channel). These shipment volumes are

Product brand splits and processor splits are also based primarily on shipments (ownership transfer) from the vendor to a third party.

Channel distribution/splits reflect a breakdown of vendor shipments by channel at the final-tier level (i.e., where PCDs are purchased by final users). Indeed, in the case of PCDs sold through a first-tier distributor, PCDs will be redistributed at the sec- -

corporate dealers and retailers. To summarize, shipment volume is based on vendor
ection of
sales out
projection.

For reexported shipments, IDC counts the channel from which the end user purchased in the country of final use (which is often different than the channel responsible for reexporting while the product is in transit).

Customer segment splits are similarly quantified and analyzed. Volume is based on vendor shipments to channels, and therefore weighted by those volumes but reflect a projection of the final purchase of the PCDs by the different types of customers. For ple, shipments registered by vendors in late September (the third quarter) may

fourth quarter). To address this and reconcile with validated vendor shipment data,

system ships to a user (the fourth quarter in this example). This matches validated

rket behavior and provide user analysis that enables IDC to correlate PCD sales with user spending and installed base information. To summarize,

reflect a projection of which the user segment is expected to purchase the PCD (a sales out projection).

IDC believes that this combined approach based on sales in, sales through, and sales out projection leads to the most accurate view of the market and, within the time and cost constraints imposed, reflects the market (rather than the industry or channel only) and is correlated with downstream market data such as installed base.

Average elling rocessor

Average selling price. The ASP is the average end-user (street) price paid for a typically configured system. Taxes that are embedded in the cost of goods (import,
- -- -sale taxes (e.g., VAT) are generally excluded. This

business investments and so forth.

Because ASPASP is an estimate of the final price by the average end users, it includes the implicit channel markup embedded in the final price. However, discounts

IDC in IDC
information available.

List price. This is the non-discounted direct price for a product. List prices are
IDC
s tracker data.

Exception: In the case of subsidized telco models where there is no other ASP

ASP.

Revenue (or hipment v

Value (i.e., end-user spending or customer revenue) represents the amount of

because many clients find spending to be a more useful figure than vendor revenue

ASP. = ASP.

s Worldwide Quarterly Personal Computing Device Tracker query tool is in U.S. dollars (typically expressed in millions [US$M]) by default, although it can be represented in a variety of other currencies as well.

Value includes all freight, insurance, and other shipping and handling fees such as taxes and tariffs that are included in vendor or channel pricing. Taxes that are added at the point of sale such as VAT or sales tax are excluded. Value represents those dollars spent by the end customer for the product as typically configured and packaged. This typically includes:

- Frame or cabinet and all cables
- Processors
- Memory
- Storage
- Communications boards
- Operating system software
- Other bundled software
- Keyboard and mouse if bundled

To a large degree, most of the aforementioned components are sold today as a bundle, with the possible exception of the OS software. IDC s revenue figures are based on street (i.e., realistic) pricing for the average model as it is assembled when the user plugs it in. OS software is included only when bundled with the initial sale and when it represents the base number of users supported by that initial license.

No other peripherals (e.g., printers and modems) are included in our ASP or value data.

Value includes revenue recognized by the hardware vendor (factory revenue items), the channel margin (when a product is sold through a third-party sales organization), and any modifications added to a system in the channel before shipping to end users that do not originate with the named hardware vendor. IDC presents data in factory revenue for some products to determine market share position on a revenue basis. IDC also presents data on customer revenue (i.e., end-user spending) to better represent the total amount of spending in the market for market sizing and forecasting purposes.

Factory revenue represents the amount of money recognized by the vendor for the sale of products. This detail is tracked for enterprise products (which typically covers a broad range of ASPs) but not for consumer device trackers where ASPs tend to remain in a relatively narrow band and where actual spending is more useful.

Installed base

The installed base reflects all systems still in use. This is generally calculated as the sum of historical shipments over an extended period, minus retirements (systems no longer in use).

Exchange ates

The time periods used in IDC trackers and forecasts are calendar time periods (quarters or years). Historical market values in the tracker for each quarter are based on the current U.S. average dollar exchange rate for that quarter as shown at **www.oanda.com**.

Therefore, for regional markets outside the United States, these as-published values are based on the corresponding average exchange rate for that country each quarter. Please refer to the IDC Tracker web query tool or regional and country research studies to find historical data for market size and growth in local currencies.

Quarterly and annual forecasts are done at a constant exchange rate at the time the forecast is developed. IDC uses the average exchange rate of the quarter or year when the - -quarter deliverable (January March), if the average exchange rate in Brazil is 2.3, then we assume that the exchange rate will remain

LEARN MORE

Methodology

In segmenting the PCD industry into its constituent markets, IDC analysts follow a methodology using demand-side and supply-side analysis of key trends and events over a

methods, and definitions.

s global market structure, methodology, and definitions are based on input from local IDC analysts in over 90 countries as well as dedicated regional analysts from IDC s regional centers in Europe, Asia/Pacific, the Middle East and Africa, Latin America, and North

the regional level and then synthesized at the worldwide level to create global market

This taxonomy is the foundation for the following:

- **s personal computing device tracker.** s Worldwide Quarterly Personal Computing Device Tracker provides broad coverage of key market information, enabling companies to compare new developments in the market from a global trends within the PCD market.
- **IDC research documents.** IDC research documents include market forecasts, competitive analysis, vendor profiles, and information on customer requirements and research projects that give in-depth analysis on specific topics.

Synopsis

This IDC study provides a comprehensive overview of and definitions for personal computing devices (PCDs). The definitions provided in this document represent the scope of s PCD hardware research.

This taxonomy outlines market segmentations and measurement methodologies for the PCD industry, including classification by technology, vendor, geography, and revenue metrics.

s worldwide PCD taxonomy presents a standardized and detailed framework of the PCD market, said Loren Loverde, program vice president, IDC s Worldwide Quarterly Personal Computing Device Tracker. The taxonomy document serves as a guideline to better understand and use IDC s PCD research.

ABOUT IDC

International Data Corporation (IDC) is the premier global market intelligence, data, and events provider for the information technology, telecommunications, and consumer technology markets. With more than 1,300 analysts worldwide, IDC offers global, regional, and local expertise on technology and industry opportunities and trends in over 110
IDC s analysis and insight help IT professionals, business executives, and the investment community make fact-based technology decisions and achieve their k
business objectives.

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